

EchoScriptor: Automatic Lifelogging Narratives via Activity-Based Audio–Language Model

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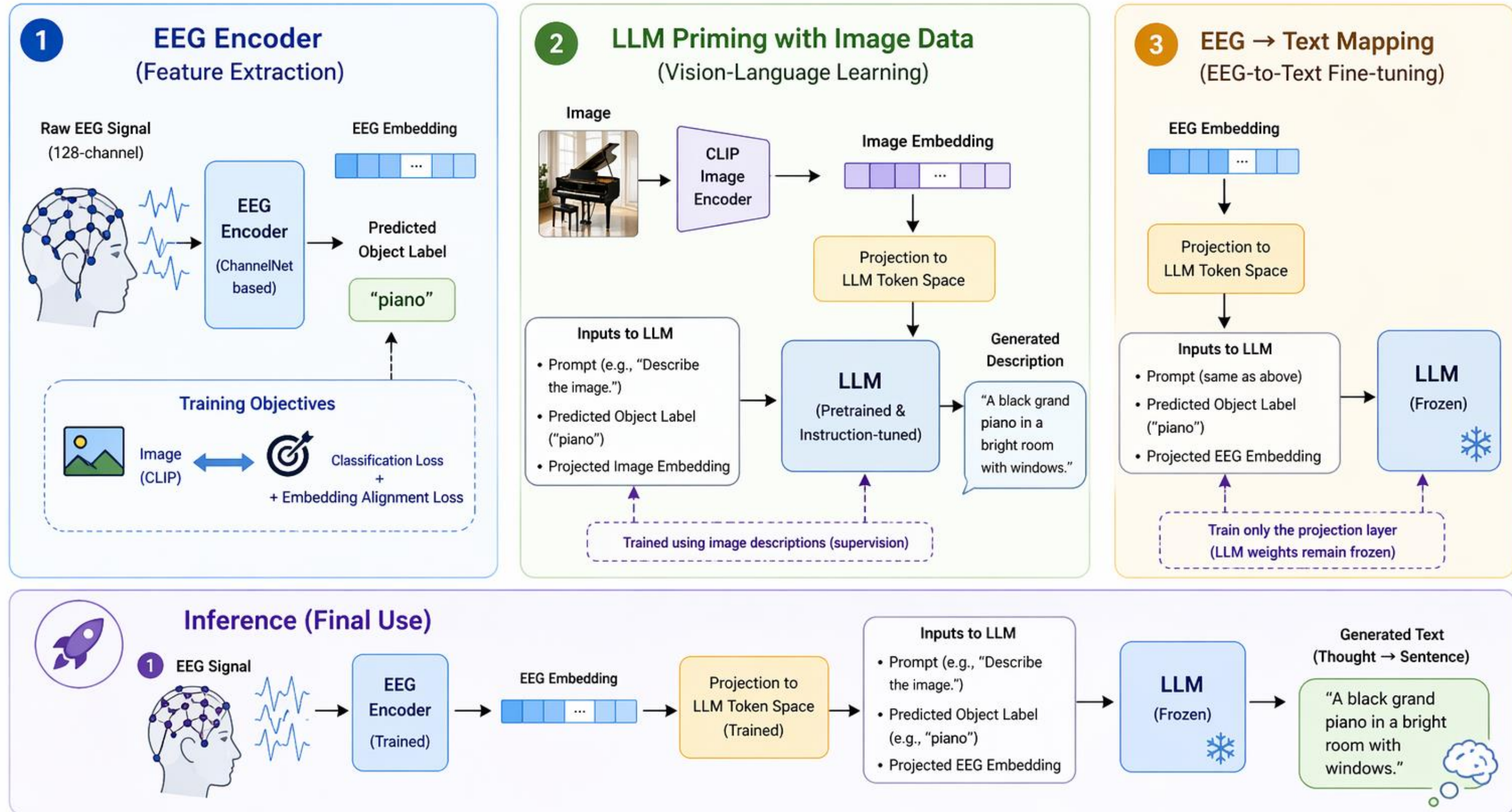
1. Thought2Text: Text Generation from EEG Signal using Large Language Models (LLMs)

2. SensorLLM: Aligning Large Language Models with Motion Sensors for Human Activity Recognition

Date: 21th April, 20256

Thought2Text: Methodology Overview

Convert EEG brain signals into meaningful text using Large Language Models (LLMs)



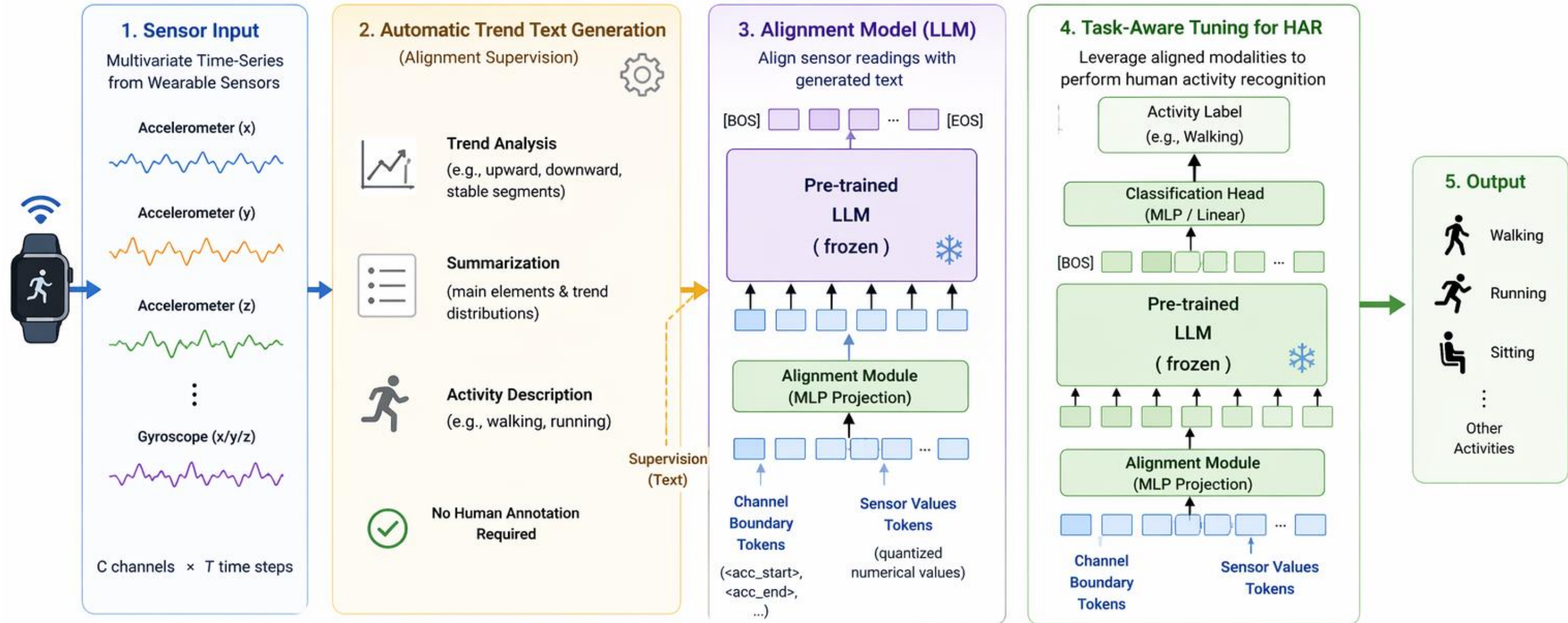
Key Idea: Bridge brain signals with vision and language models through a shared embedding space to generate natural language.

SensorLLM: Methodology Overview

Aligning Large Language Models with Motion Sensors for Human Activity Recognition

Stage 1: Sensor-Language Alignment (Pre-training)

Stage 2: Task-Aware Tuning (Fine-tuning)



Key Idea

SensorLLM first aligns sensor time-series with human-intuitive trend descriptions using automatically generated text. Then it leverages the aligned knowledge to perform Human Activity Recognition.



Benefits

- No manual annotation required
- Handles multi-channel & long sequences
- Generalizes across datasets and sensors



Frozen LLM



Trainable Module



Input Tokens



Text Tokens



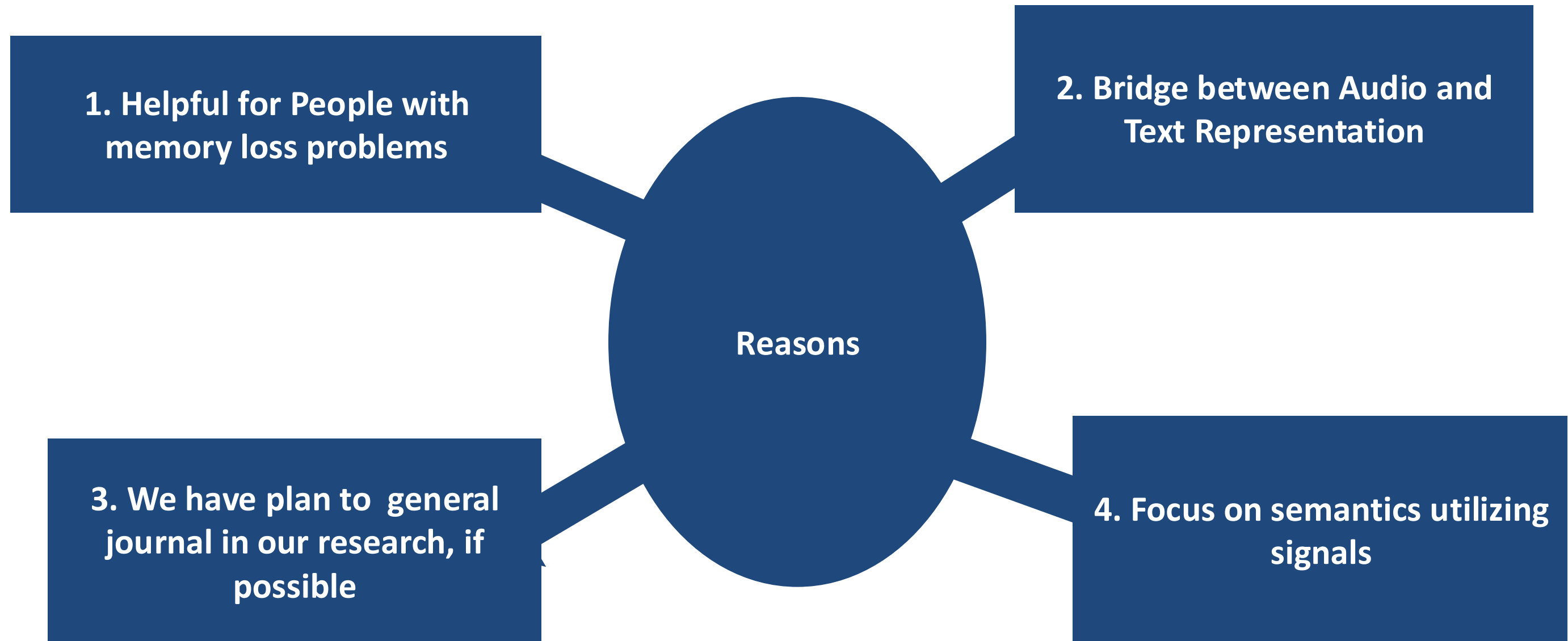
Output / Label

HAR = Human Activity Recognition

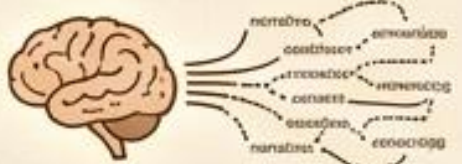
Overview of Discussion

1. Reasons for Choosing the Paper
2. Gap of Work So far
3. Dataset Construction
4. Methodology
5. Evaluation
6. Thoughts
7. Conclusion

Why This Paper?

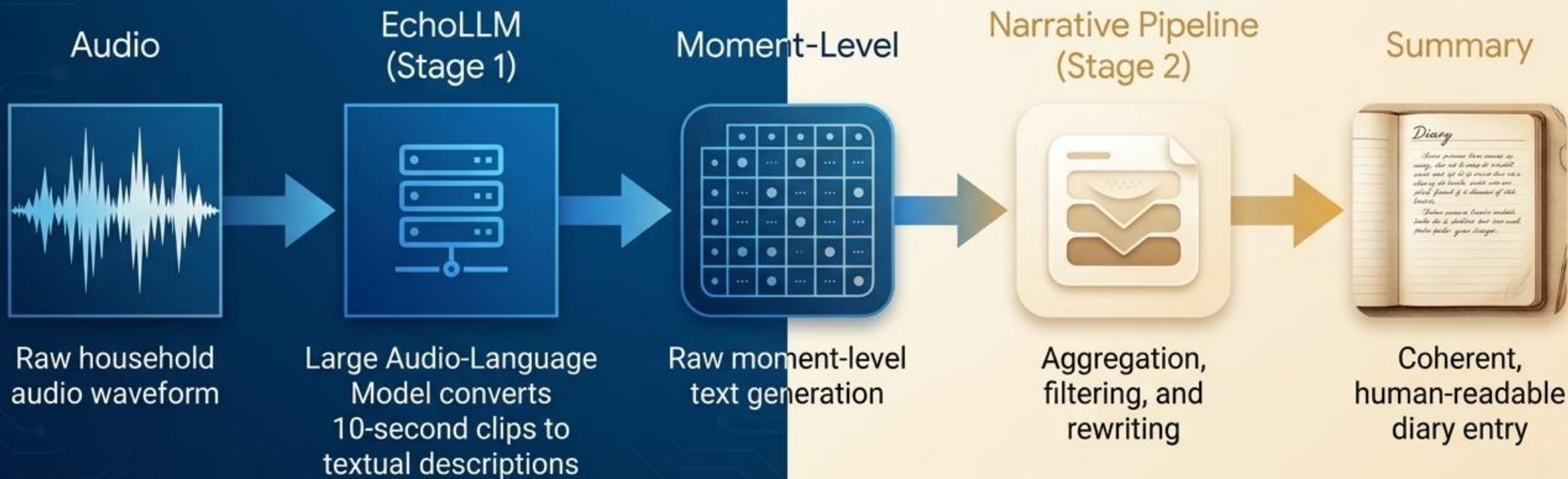


The Gap in Unobtrusive Memory Support

 Wearable Cameras	 IMU / GPS	 Audio Classification	 EchoScriptor
 Captures rich context, but severely compromised by privacy concerns and requires manual annotation. 	Highly passive and private, but suffers from a narrow scope limited to locomotion and sleep. Cannot capture complex domestic tasks. 	 Passive and private, but lacks context. Reduces life to sterile, discrete closed-set labels like Vacuum or Door. 	 The ideal state. Passive, private, highly expressive, and deeply context-aware using Audio-Language Generation . 

Lifelogging for cognitive care requires continuous, unobtrusive semantic understanding, not just discrete event labels.

The Journey from Raw Signal to Semantic Story



How 463,532 Training Clips Were Built from Scratch

EchoScriptor · Dataset Construction — 5 Steps



Final Total: $199,800 + \sim 161,550 + \sim 97,800 + 4,382 = 463,532$ clips

Collect Raw Recordings

Activity Recordings

313

Long Audio Files

- 24 household activity types
- 10–13 recordings per activity
- Examples: hairdryer, keyboard typing, vacuum cleaner, baby crying
- Source: YouTube & Freesound

Background Recordings

80

Long Audio Files

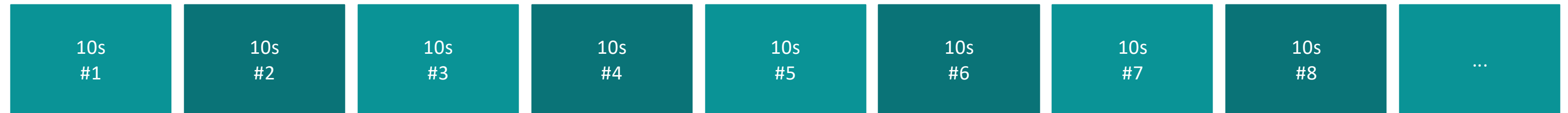
- 8 common home background types
- 10 recordings per background
- Examples: air conditioning, traffic noise, fridge humming, birds chirping
- Source: YouTube & Freesound

Chop into 10-Second Clips

Each long recording is sliced into non-overlapping 10-second segments:

Long Raw Recording (~20 minutes) [55 % Training for background and 90% training for Activity]

▼ slice into 10s segments



Activity Clips (per type)

~27

training clips
per activity type

9:1 split → ~27 train, ~3 test per class

Background Clips (per type)

~10

training clips
per background type

Adjusted split → ~10 train, 6–8 test per class

Total Initial Pool

$648 + 80 ((24 * 27) + 10 * 8)$

pure clips total
(before augmentation)

Very small — augmentation needed in Step 4

Augment Pure Clips

Why? The model needs to handle real-world cases where ONLY an activity OR ONLY background is present — not just mixed audio.

Augmentation Process

1. Slice long recordings into many non-overlapping 10s segments
→ captures natural acoustic variation over time

2. Apply transformations to each segment:

• Volume scaling (louder / softer)

• Frequency equalization (brighter / darker sound)

3. Each transformed segment = a new unique training clip

Before vs. After Augmentation

Type	Before	After
Activity clips	648	~161,550 ×249
Background clips	80	~97,800 ×1,222

💡

Without augmentation, the model would have only $648+80 = 728$ pure clips — far too few compared to 199,800 mixed clips. Augmentation balances the training set so the model learns all conditions equally well.

Add White Noise (Null Condition)

What is it?

4,382 artificial audio clips

containing only white noise — no activity, no background.

Ground truth label:

"There is no meaningful home event happening."

Split:

4,382 training clips + 486 test clips

Why is it needed?

Without white noise training:

The model might hallucinate an activity even when the audio is completely silent or irrelevant.

With white noise training:

The model learns to output "nothing is happening" when it should — making it robust and reliable in real-world conditions.

Result: 100% accuracy on noise-only test clips 

4,382 training + 486 test = 4,868 white noise clips total

Final Training Dataset

Step 3

Activity + Background Mixtures

199,800

Step 4a

Augmented Pure Activity Clips

~161,550

Step 4b

Augmented Pure Background Clips

~97,800

Step 5

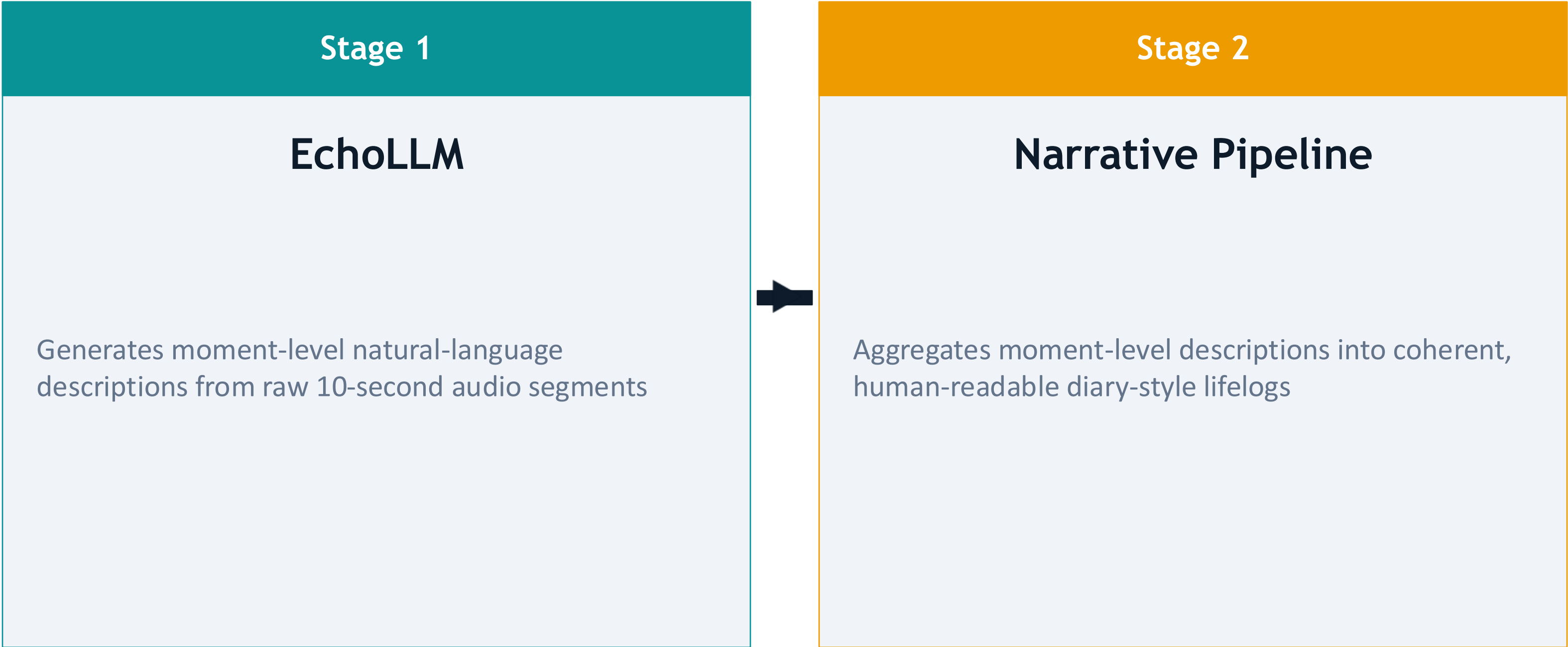
White Noise (Null Condition)

4,382

TOTAL

463,532 training clips

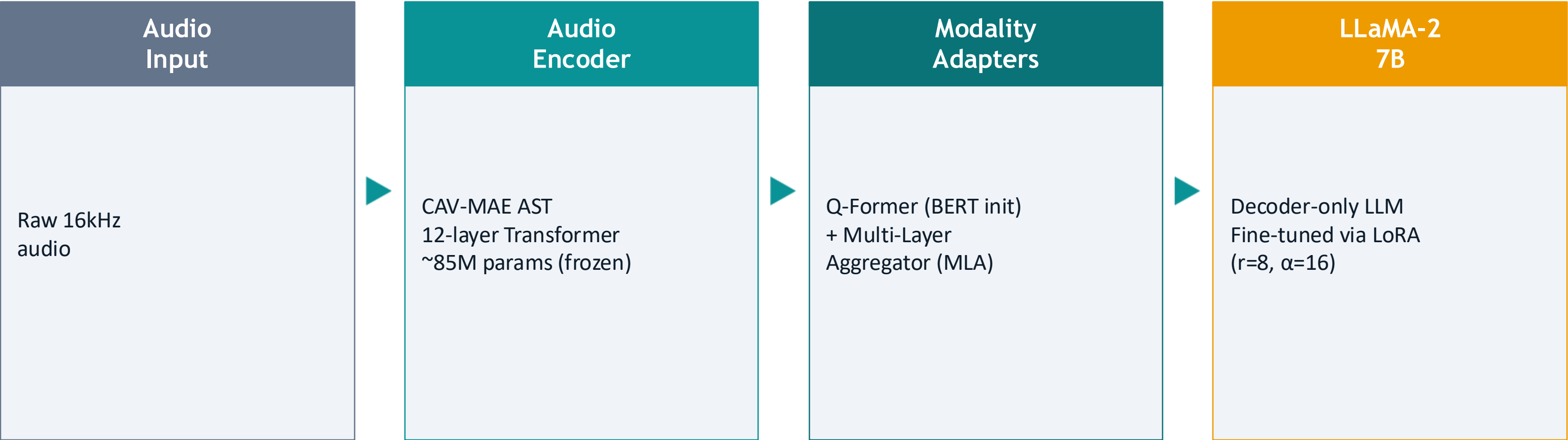
System Overview



 Raw Household Audio

 Narrative Lifelog Summary

Stage 1: EchoLLM Architecture



Output: Moment-Level Description

"The user is drying hair with a hairdryer in the bathroom, accompanied by the low hum of the air conditioner."

Audio Encoder: CAV-MAE / AST

Architecture

Audio Spectrogram Transformer (AST) — 12-layer Transformer, hidden size 768, 12 attention heads, $\approx 85\text{M}$ parameters

Pre-training

Self-supervised masked autoencoding on AudioSet-2M and VGGSound via CAV-MAE objective — learns robust, semantically rich representations

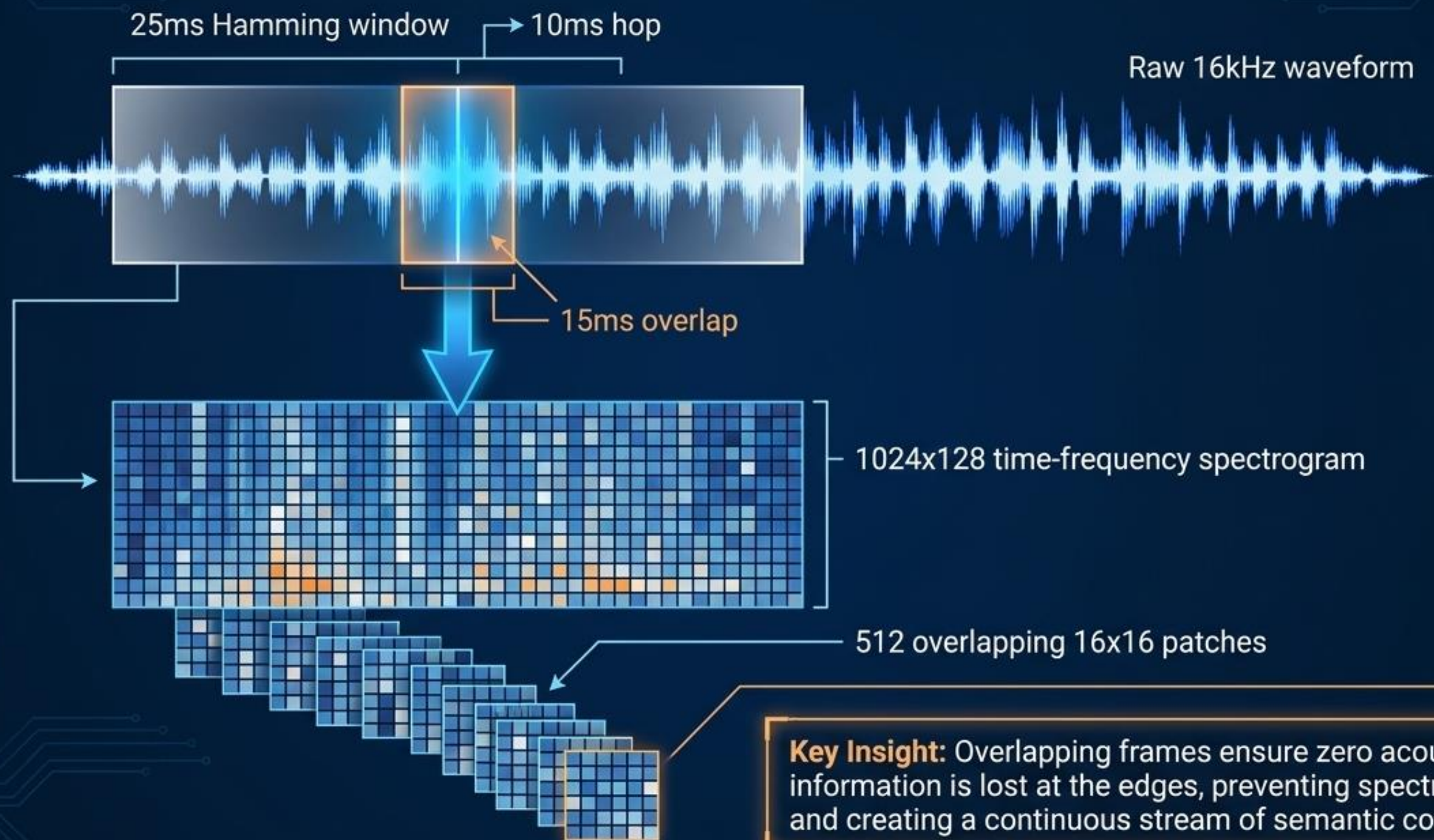
Tokenization

Each 10-second audio $\rightarrow$ 128-bin log-Mel spectrogram (1024×128) $\rightarrow$ split into 512 overlapping 16×16 patches, each projected to 768-dim embedding

Frozen During Training

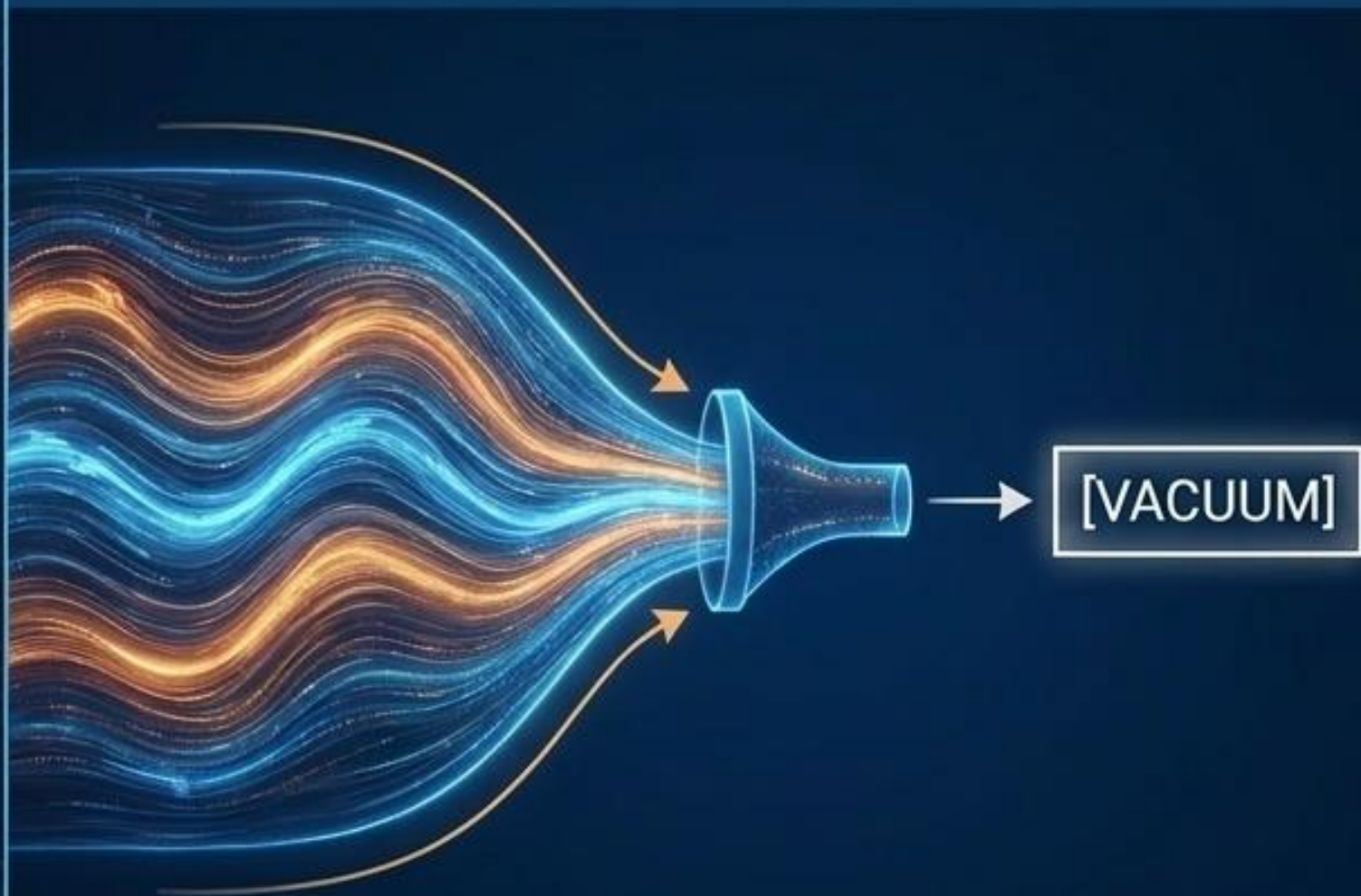
All AST parameters are frozen — preserves general pre-trained knowledge; optimization focuses entirely on the modality adapter and LLM

Slicing Sound for Continuous Understanding



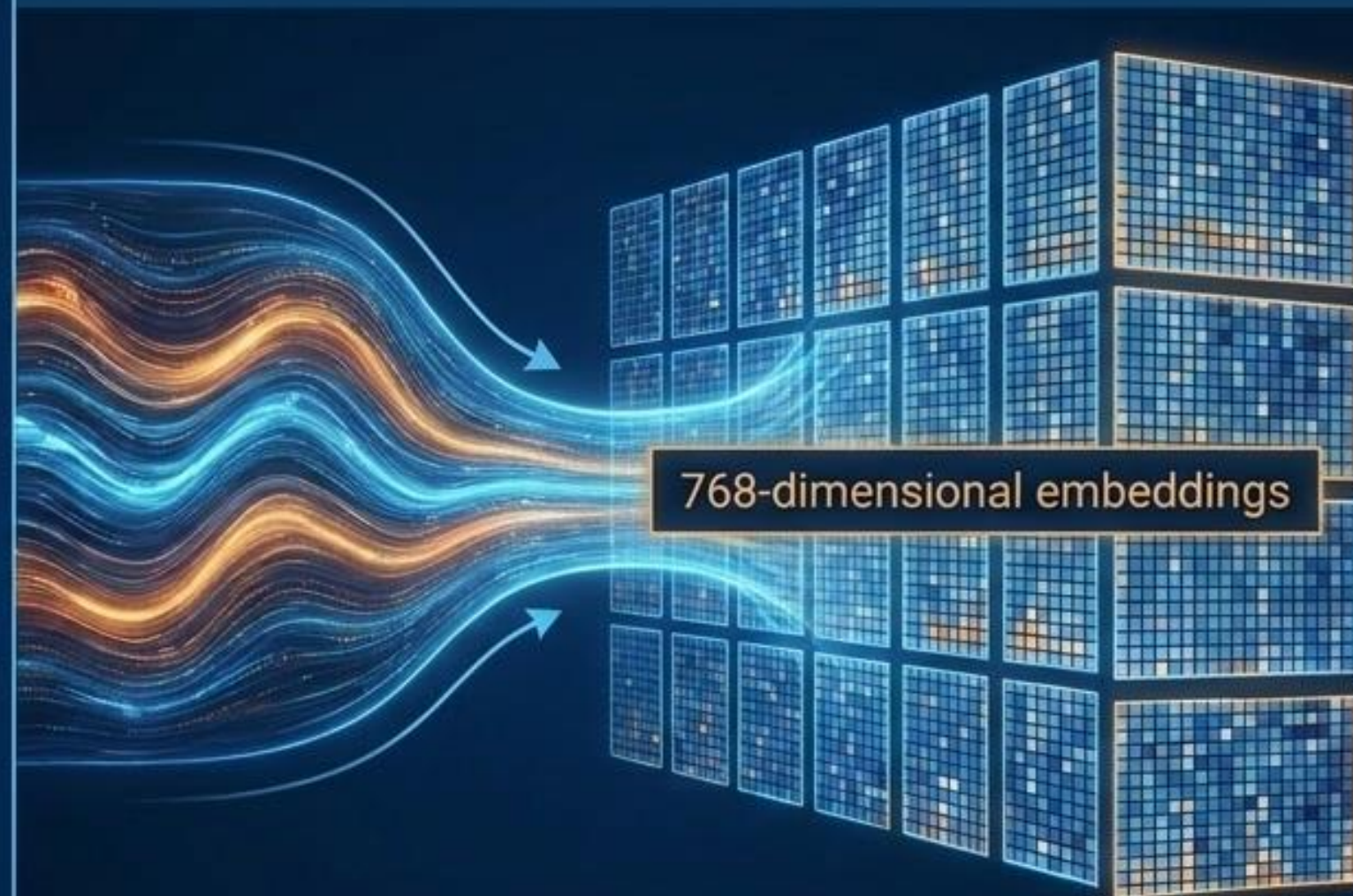
Removing the Bottleneck of Discrete Labels

Standard AST (Classification Head)



Destroys overlapping sounds and subtle acoustic context.

EchoScriptor (Frozen AST + Token Features)



Rich, continuous data flow preserving both the primary action and the background environment. Extracts 85M parameters of generalized auditory knowledge without restricting the output taxonomy.

Modality Adapters: Q-Former & MLA

Audio Q-Former

Based on: Querying Transformer (BLIP-2)

Init from: BERT weights

Input: Final-layer AST features

Training: Audio-text matching, grounded text generation, contrastive objectives

Purpose: Projects audio into LLM embedding space (semantic bridging)

Multi-Layer Aggregator (MLA)

Architecture: Transformer-style with self- and cross-attention

Input: Multiple intermediate AST layers + final layer

Key Insight: Different AST layers capture low-level spectral textures → high-level acoustic concepts

Purpose: Fuses multi-scale audio signals for holistic, fine-grained representation

LLM Fine-tuning with LoRA

LLaMA-2 (7B)

- Decoder-only autoregressive Transformer
- Pre-trained on large-scale text corpora
- Receives aligned audio embeddings + text instruction prompt
- Autoregressively generates natural-language activity description

LoRA Configuration

- Rank $r = 8$, $\alpha = 16$, dropout $p = 0.05$
- Trained for 5 epochs on 4× NVIDIA RTX 6000 Ada
- Preserves pre-trained GAMA knowledge while reducing GPU cost
- Masked cross-entropy loss — question tokens masked, only answer tokens optimized

Masked Cross-Entropy Loss:

$$L_{\text{masked}} = -\sum m_t \cdot \log \hat{y}_{\{t, y_t\}} \quad \text{where } m_t = 0 \text{ (question tokens)} \mid m_t = 1 \text{ (answer tokens)}$$

Teaching a Language Model to Hear

Input: 768-dimensional audio embeddings from the frozen AST.

Audio Q-Former

Projects final-layer features into rich semantic representations.

Multi-Layer Aggregator (MLA)

Fuses intermediate layers to capture low-level textures and high-level concepts.

LLM Embedding Layer

Takeaway: Text tokens and audio embeddings exist in completely different mathematical spaces. The Q-Former and MLA act as universal translators, aligning sound features into a language LLaMA 2-7B can natively process.

The Architecture is the Empathy Engine



The Tech: Preserving 128-bin spectrograms and aligning them via MLA allows the model to process overlapping, multi-layered audio simultaneously.



The Human Benefit: A lifelogging system remembers not just that a patient washed hands, but that they rinsed a mug while the air conditioner hummed comfortably.

Stage 2: Narrative Summary

Construction Pipeline

Turning 10-second moment descriptions into coherent human-readable diary entries



All 3 steps use GPT-4o · Input: raw moment-level descriptions · Output: diary-style lifelog

The Problem

EchoLLM outputs one description every 10 seconds. A 20-minute recording produces 120 raw descriptions like these:

Raw EchoLLM Output (every 10s)

1:30–1:40	The sound of a hair dryer fills the bathroom	✓
1:40–1:50	The user is styling hair with a hair dryer in the bathroom	✓
1:50–2:00	The user is using a blender	✗
2:00–2:10	The user is using a hairdryer with humming sound	✓
2:10–2:20	The user is drying hair with a hairdryer	✓
6:00–6:10	The sound of mechanical keyboard fills the room	✓
6:10–6:20	The sound of typing keyboard clacked rhythmically	✓
6:20–6:30	The user focuses on work while typing on the keyboard	✓

3 Problems

① Noise

Single wrong prediction (blender at 1:50) breaks the flow

② Repetition

Same activity described 4–5 times in slightly different words

③ No narrative

120 separate sentences ≠ a readable diary entry

Solution → Run through the 5-step Narrative Construction Pipeline

1

Semantic Filtering — Remove the Noise

Core Idea:

Human activities are continuous. If something appears only once or twice in a row, it's probably a misclassification — not a real event. GPT-4o applies rules to remove these brief anomalies.

The 3 Filtering Rules:

Continuous Sounds

e.g. hairdryer, keyboard, chopping, vacuum

Rule: Must appear in ≥ 3 consecutive segments

These are sustained activities — if they appear for less than 30 seconds straight, it's likely a false detection

Faint Background Sounds

e.g. fridge hum, air conditioning, rain

Rule: Must appear in ≥ 4 consecutive segments

Background sounds are subtle and easy to confuse — need stronger evidence before accepting them as real

Short Events

e.g. toothbrush, talking, toilet flush

Rule: Must appear in ≥ 2 consecutive segments

These are brief by nature but should still appear at least twice back-to-back to be considered real

✗ Removed: "blender" appearing only once at 1:50 between a sequence of hairdryer detections → treated as misclassification

1 Semantic Filtering – Before vs After

BEFORE Filtering		
1:30–1:40	hair dryer fills the bathroom	✓
1:40–1:50	styling hair with hair dryer	✓
1:50–2:00	user is using a blender	✗ REMOVED
2:00–2:10	hairdryer with humming sound	✓
2:10–2:20	drying hair with a hairdryer	✓
2:20–2:30	drying hair with a hairdryer	✓
6:00–6:10	mechanical keyboard fills the room	✓
6:10–6:20	typing keyboard clacked rhythmically	✓
6:20–6:30	user focuses on work, typing keyboard	✓



AFTER Filtering	
1:30–1:40	hair dryer fills the bathroom
1:40–1:50	styling hair with hair dryer
2:00–2:10	hairdryer with humming sound
2:10–2:20	drying hair with a hairdryer
2:20–2:30	drying hair with a hairdryer
6:00–6:10	mechanical keyboard fills the room
6:10–6:20	typing keyboard clacked rhythmically
6:20–6:30	user focuses on work, typing keyboard

Result: 1 noisy prediction removed. Clean, consistent sequence ready for Step 2.

The Semantic Low-Pass Filter



Context Rules:

- **Continuous sounds:** Must appear in at least 3 consecutive segments.
- **Background sounds:** Must persist across at least 4 consecutive segments.
- **Short activities:** Must occur at least 2 times consecutively.

2

Activity Block Construction – Group Related Events

Core Idea:

After filtering, we still have many segments that all describe the same activity. GPT-4o collapses consecutive related segments into a single unified "activity block" — one label for the whole run.

Many segments → One block:

hair dryer fills bathroom

styling hair with hairdryer

hairdryer with humming
sound

drying hair with a hairdryer

drying hair with a hairdryer

▼ GPT-4o collapses into one block

● the user is drying hair with a hairdryer

How GPT-4o decides what to group:

✓ Temporal continuity

Consecutive segments describing the same or very similar activity → merged into one block

✓ Spatial continuity

Same location context (e.g. 'in bathroom') → kept together as one unified event

✗ Outlier removal

Any mislabeled slice within a run (e.g. 'chopping' inside a hairdryer sequence) → flagged and discarded

3

Narrative Summarization — Write the Diary Entry

Core Idea:

GPT-4o takes the cleaned activity blocks and rewrites them into a single fluent, third-person, chronologically ordered paragraph — like a diary entry written about someone's day.

Input: Activity Blocks

1:30 – 2:30

drying hair with hairdryer

6:00 – 6:30

typing on the keyboard

9:10 – 9:40

running water in the sink

12:00 – 12:50

playing piano

GPT-4o
rewriting
→

Output: Narrative Lifelog

"The user started the day with the sound of a hair dryer humming in the bathroom.

This was followed by a session of focused work, as the mechanical keyboard clacked rhythmically in the room.

Later, the gentle sound of running water suggested a moment of handwashing or cleaning.

The day concluded with the melodic sound of piano filling the room."



Chronological: Events appear in the order they happened



Third-person: Always written as 'the user...'



Fluent prose: Reads like a real diary — not a list

End-to-End: Raw Audio → Lifelog

EchoLLM

1 description every 10s

"The user is drying hair with a hairdryer in the bathroom, accompanied by the low hum of the air conditioner."

Step 1 Semantic Filter

Remove noisy/brief predictions

Keeps only consistent runs of $\geq 2-4$ consecutive segments. Removes the one 'blender' prediction.

Step 2 Block Construction

Collapse repeats into blocks

5 hairdryer segments → one block: 'user is drying hair with a hairdryer' (1:30–2:30)

Step 3 Narrative Summary

Rewrite into diary prose

"The user started the day with the sound of a hair dryer humming in the bathroom..."

Crafting the Human Narrative

Block 1: [1:30 - 2:20]
User is drying hair
with a hairdryer.

Block 2: [6:00 - 6:30]
User is typing on the
keyboard.

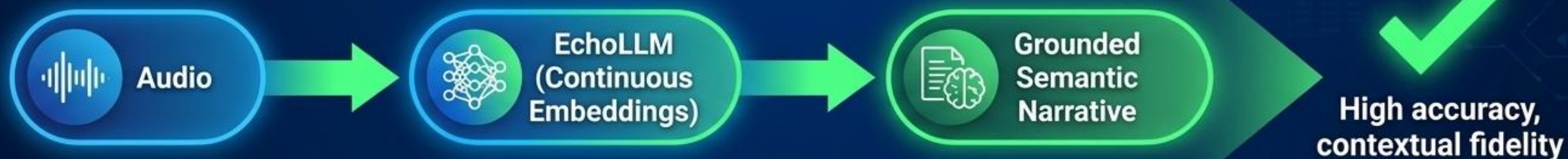


The user started the day with the sound of a hair dryer humming in the bathroom. This was followed by a session of focused work, as the mechanical keyboard clacked rhythmically in the room.

GPT-4o rewrites chronological activity blocks into a concise **third-person** account, creating an accessible, coherent artifact for memory recall.

Why Direct Generation Beats Discrete Labels

EchoScriptor



Baseline: AST + LLM



Takeaway: Reducing rich audio to text labels before generating a summary destroys the acoustic reality. EchoScriptor keeps the audio alive until the final generation step.

Two Evaluation Protocols

Protocol 1 — LLM-as-a-Judge

GPT-4o acts as a human examiner, scoring each generated description against the ground truth on a 0–5 scale:

0 Completely unrelated
(hairdryer vs vacuum)

1 Minimal relation
(toothbrush vs shaver)

2 Loose relation
(microwave vs coffee)

3 Same category, different act
(shower vs face wash)

4 Closely related
(shower vs running water)

5 Identical meaning ✓
(vacuum vs vacuum cleaner)

Evaluated on: Activity Accuracy · Background Accuracy · Overall Expression Quality

Protocol 2 — NLP Similarity Metrics

Automatic, objective metrics comparing generated text against ground-truth descriptions:

BERTScore-F1

Uses contextual embeddings from BERT to measure how semantically similar the generated description is to the reference — captures meaning, not just word overlap.

Sentence-BERT (Cosine Similarity)

Encodes full sentences into embedding vectors and measures cosine similarity — evaluates sentence-level alignment and overall coherence of the description.

Evaluation Results

94.15%

Activity Recognition
Accuracy

89.25%

Background Recognition
Accuracy

F1 = 0.92

Summary-Level
F1 Score

EchoLLM vs. AST+LLM Baseline

Metric	AST+LLM	EchoLLM	Δ
Activity Accuracy (Prop=5)	86.72%	94.15%	+7.43pp
Background Accuracy (Prop=5)	83.93%	89.25%	+5.32pp
Overall Expression (Mean)	4.22/5	4.72/5	+0.50
BERTScore-F1	0.8846	0.9075	+0.023
Sentence-BERT	0.5942	0.7030	+0.109

Results on Pure & Noise Conditions

Pure Activity & Background Results

Condition	Source-Type Detection	Description Accuracy
Pure Activity (106 clips)	94.34%	99.06%
Pure Background (213 clips)	100.00%	98.13%
Weighted Average	98.12%	98.43%

Key Insight: EchoLLM can reliably distinguish activity-only vs background-only vs overlapping — something traditional classifiers cannot do since they only output a ranked list of labels.

White Noise Results

100%

Accuracy on 486 noise-only clips

The model correctly said "nothing is happening" for every single noise clip — zero hallucinations.

Real-World Validation (3 participants, own homes, chest-mounted smartphone):

Average activity recognition accuracy: 88.53% (range: 84.1% – 92.0%) — without any post-processing or filtering

Narrative Summary – Quantitative Evaluation Setup

Data Collection

- 10 participants, 10 distinct households
- Each performed routine activities naturally at home
- Recorded via chest-mounted iPhone 13 Pro
- 10–20 minutes per session
- Varied room layouts, devices, time of day

Each recording produced 3 summaries:

- ① EchoScriptor
- ② AST+LLM+GPT baseline
- ③ Human-written (by the participant)

3 Categories of Metrics

① Classification Accuracy

Precision · Recall · F1-score

Did the summary mention the right activities?
(ignores order)

② Semantic Similarity

BERTScore-F1 · Sentence-BERT

Are the descriptions semantically faithful
to what actually happened?

③ Activity Alignment

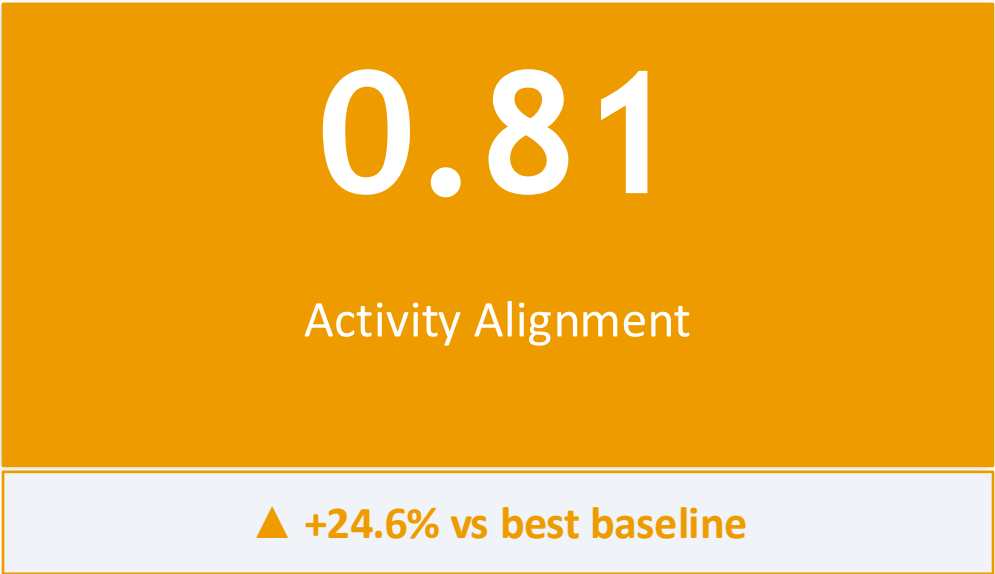
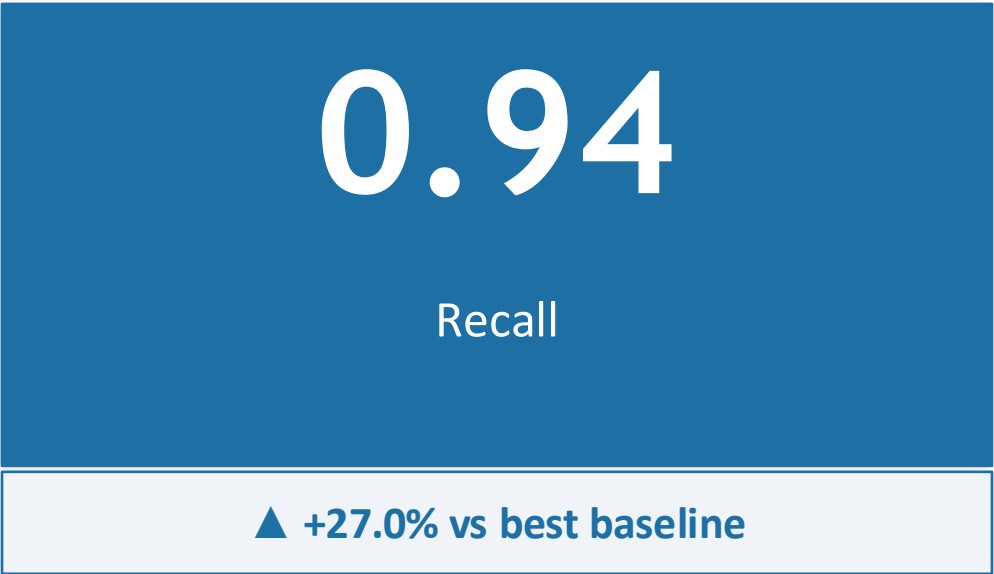
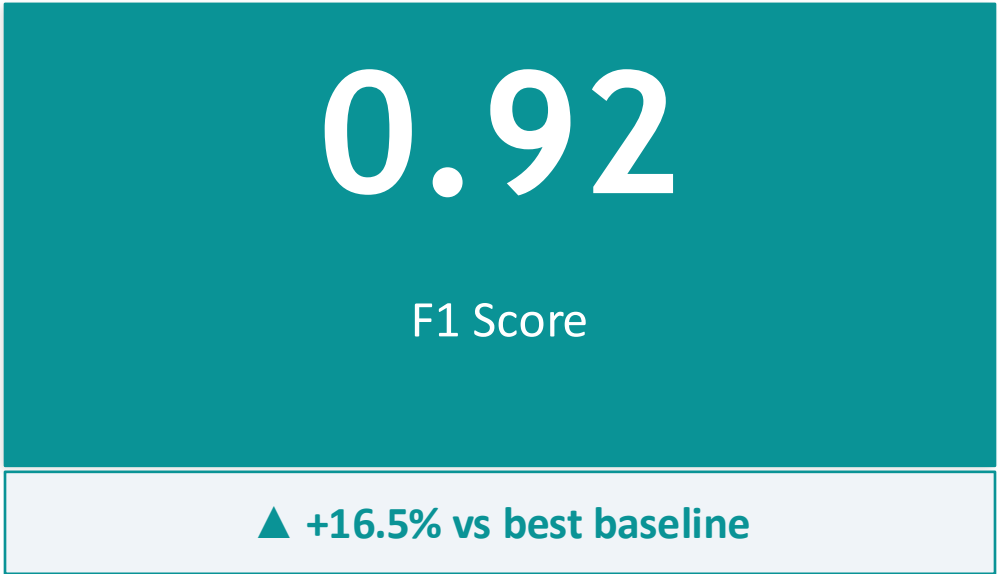
Normalized Levenshtein edit distance

Did the summary list activities in the CORRECT ORDER?
(order matters for lifelogging!)

Alignment Formula:

$Align(P, G) = 1 - EditDist(P, G) / \max(|P|, |G|)$ where P = predicted sequence, G = ground truth sequence

Narrative Summary – Quantitative Results



EchoScriptor vs All Baseline Methods

Metric	EchoScriptor	Method 1 AST+LLM+GPT	Method 2 AST+LLM+GPT	Method 3 AST+LLM+GPT
Precision	0.92 ✓	0.71	0.85	0.90
Recall	0.94 ✓	0.49	0.74	0.68
F1-Score	0.92 ✓	0.62	0.79	0.76
BERTScore	0.88 ✓	0.87	0.86	0.87
Sentence-BERT	0.68 ✓	0.56	0.55	0.59
Activity Alignment	0.81 ✓	0.46	0.65	0.59

User Perception Study – Design

Participants

20 participants
8 female, 12 male
Mean age: 24.95 (SD=3.67)
All fluent in English
University community

Each reviewed 4 of 10 videos
→ 8 independent ratings per video

Interface

Custom web app accessed via Zoom

- 2-min highlight video
(or switch to full recording)
- 3 blinded summary cards
(A, B, C — randomized order)
- Authorship hidden from participants
- All texts in 3rd-person
("the user...")

Questionnaire (11 Qs)

Q1–Q4: Summary Quality
Accuracy · Precision
Clarity · Chronological Order

Q5–Q6: Lifelogging Value
Trustworthiness · Utility

Q7–Q9: Direct Rankings
Clarity · Helpfulness · Comfort

Q10: General perceived value
Q11: Open-ended (why?)

Q1–Q6 & Q10: 7-point Likert scale (1 = strongly disagree → 7 = strongly agree)
Q7–Q9: Rank-order items (1 = best, 3 = worst) | Analysis: Linear Mixed-Effects Models + Wilcoxon signed-rank tests

Bias Controls:

- Randomized per participant
- Video contributors did NOT evaluate their own recordings
- Participants told not to let minor factual errors disproportionately influence unrelated quality dimensions

Results Summary

Stage 1

 EchoLLM: 94.15% activity accuracy, 89.25% background accuracy — outperforms AST+LLM baseline on every metric

Stage 1

 100% accuracy on white noise clips — zero hallucinations when nothing is happening

Stage 1

 Real-world in-home validation: 88.53% accuracy without any post-processing

Stage 2A

 Summary F1 = 0.92, Recall = 0.94, Alignment = 0.81 — best baseline achieves F1 = 0.79 (+16.5% improvement)

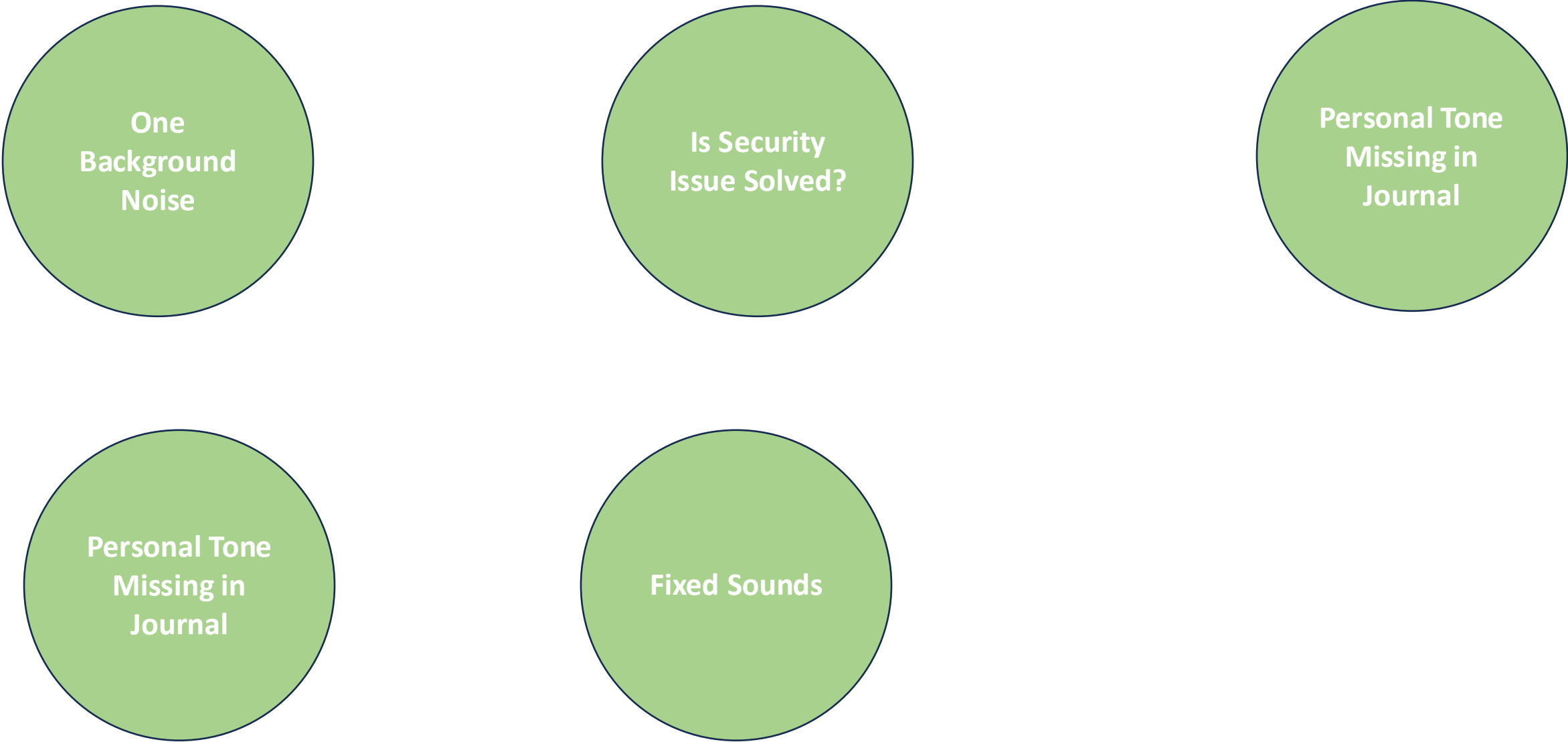
Stage 2B

 Users rated EchoScriptor (mean 5.42/7) far above baseline (3.44/7), approaching human-written quality (6.11/7)

Stage 2B

 Perceived utility (Q6): EchoScriptor not significantly different from human — practical parity achieved

Limitations



One
Background
Noise

Is Security
Issue Solved?

Personal Tone
Missing in
Journal

Personal Tone
Missing in
Journal

Fixed Sounds

